

Linear Algebra Concepts - Detailed Explanation

1. Norms and Inner Products

Norm:

A norm measures the length or magnitude of a vector.

For vector $v = (v_1, v_2, \dots, v_n)$,

$$\|v\| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$$

Properties of Norm:

- Non-negative: $\|v\| \geq 0$
- $\|v\| = 0$ only if v is zero vector
- Scalar multiplication: $\|cv\| = |c| \|v\|$
- Triangle inequality: $\|u + v\| \leq \|u\| + \|v\|$

Distance Between Vectors:

$$\text{Distance}(u, v) = \|u - v\|$$

Dot Product (Inner Product):

$$u \cdot v = u_1v_1 + u_2v_2 + \dots + u_nv_n$$

$$u \cdot v = \|u\| \|v\| \cos\theta$$

Orthogonality:

Two vectors are orthogonal if $u \cdot v = 0$

2. Orthonormality

Unit Vector:

A vector with length 1.

$$\mathbf{v}^\bullet = \mathbf{v} / \|\mathbf{v}\|$$

Orthonormal Basis:

- Each vector has norm = 1
- Vectors are mutually orthogonal
- They span the entire space

Example in $\mathbb{R}^3$:

$$\mathbf{i} = (1, 0, 0)$$

$$\mathbf{j} = (0, 1, 0)$$

$$\mathbf{k} = (0, 0, 1)$$

3. Eigenvalues and Eigenvectors

Definition:

For square matrix A ,

$$A\mathbf{v} = \lambda\mathbf{v}$$

λ is eigenvalue, $\mathbf{v}$ is eigenvector.

Characteristic Equation:

$$\det(A - \lambda I) = 0$$

For 2×2 matrix:

$$\lambda^2 - (\text{Trace})\lambda + (\text{Determinant}) = 0$$

Importance:

- Used in PCA
- Stability analysis
- Vibrations
- Machine Learning applications

4. Orthogonal Matrices

A matrix Q is orthogonal if:

$$Q^T Q = I$$

Property:

$$Q^{-1} = Q^T$$

Orthogonal matrices preserve:

- Length
- Angles
- Represent rotations/reflections

Example rotation matrix:

$$\begin{bmatrix} \cos\theta & -\sin\theta \end{bmatrix}$$

$$\begin{bmatrix} \sin\theta & \cos\theta \end{bmatrix}$$

5. Diagonalization

Definition:

$$A = P D P^{-1}$$

P contains eigenvectors.

D is diagonal matrix of eigenvalues.

Why Diagonalize?

- Determinant = product of diagonal elements
- Powers: $A^n = P D^n P^{-1}$
- Inverse: reciprocal of diagonal entries

Condition:

Matrix must have enough independent eigenvectors.